

J1 / external 24 V
USB: interface bias only

F1 / Q500 / U12
Fuse, reverse block, eFuse

V24_BUS + bulk
U501 pulse clamp

F4–F10 -> 7 ambient zones
21 MOSFET sinks / common +24 V

U14 -> V24_SPOT -> 3 x AL8860
U500 clamp / three paired LED outputs

F2/F3 -> 2 CAN motors
J16 external brake / bus port

DATA / CONTROL
GPIO2/3 I2C -> U2 0x40 / U3 0x41 -> 21 PWM sinks
GPIO10/11/18 -> 3 spotlight PWM; GPIO6/7 -> CAN
GPIO0 AND ARM AND READY -> SAFE
TMP112 0x48; J18 GPIO19-23 / 3V3
Strap probes: GPIO4/5/15; USB12/13; UART16/17

Prototype only: source/current/thermal/Rf and motor-energy tests remain pending.
Open ARM = lighting inhibited. Motor stopping/holding is a separate function.

controller

power_sequence

can

usb

pwm

zone_1

zone_2

zone_3

zone_4

zone_5

zone_6

zone_7

test

power_1

power_2

spot_1

spot_2

monitor_expansion

ENGINEERING DRAFT – see release status before fabrication

Engineered Lighting

Sheet: /
File: engineered-lighting-rev-a.kicad_sch

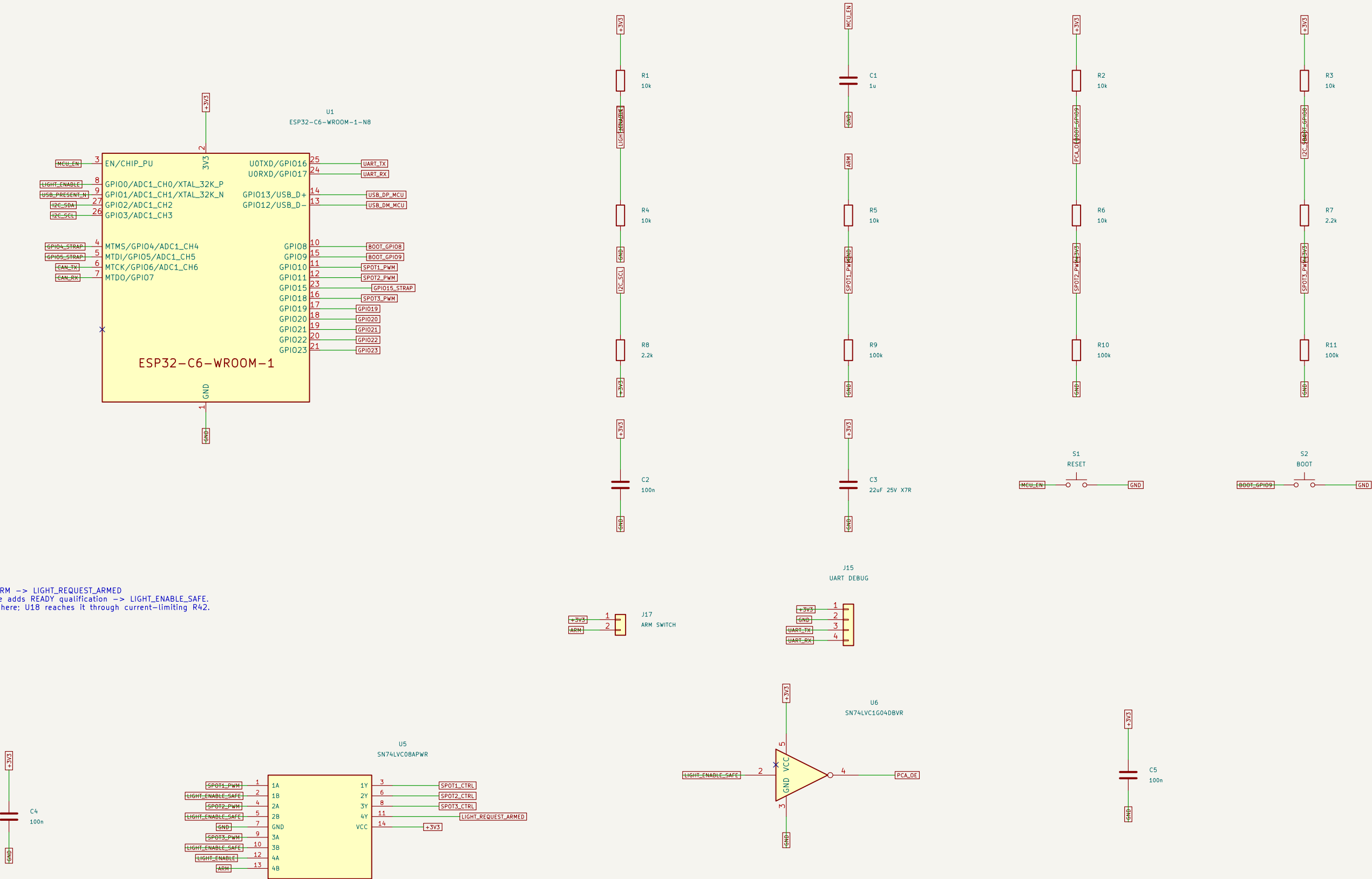
Title: Integrated lighting controller

Size: A3Date: 2026-09-06Rev: A draft

KiCad E.D.A. 10.0.3Id: 1/19

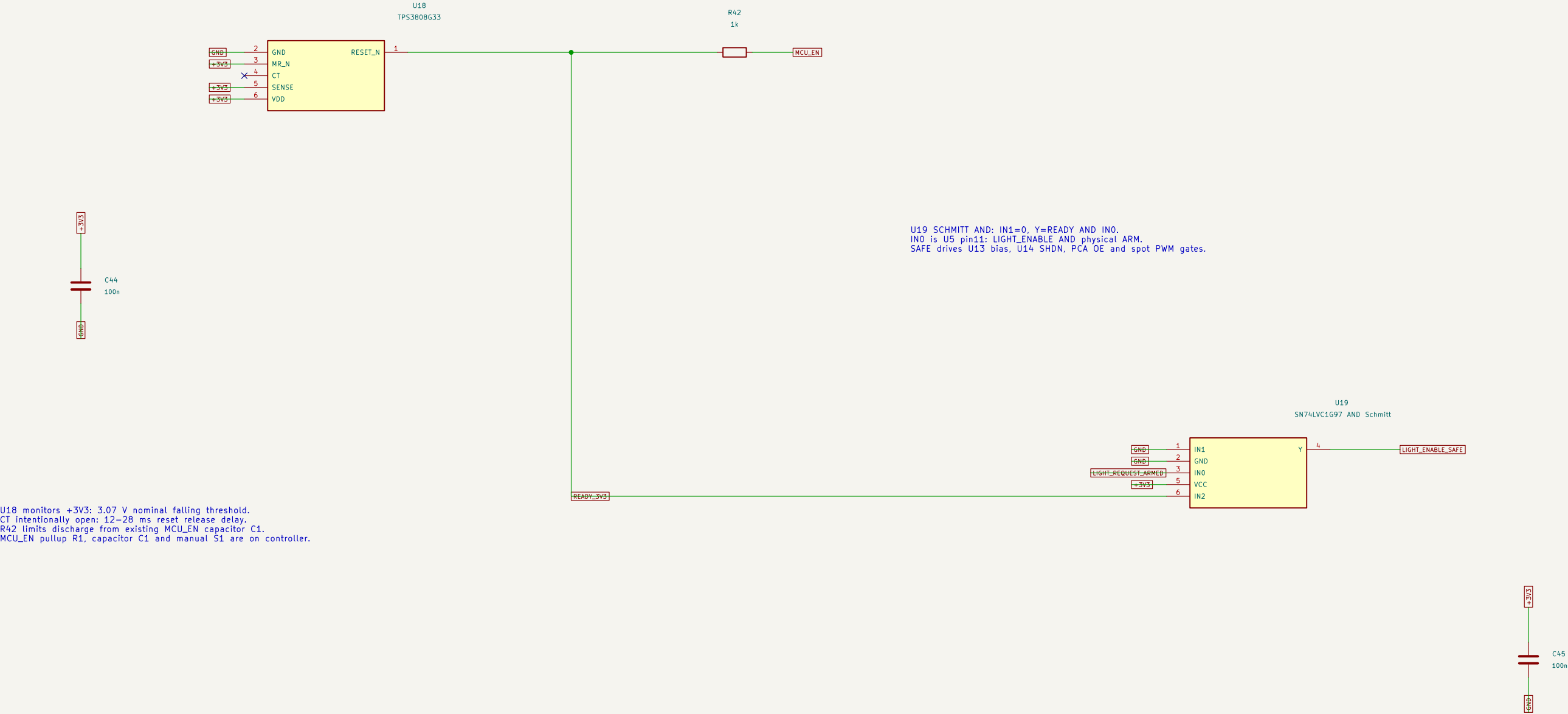
CONTROLLER

GPIO 0 AND physical ARM -> LIGHT_REQUEST_ARMED
U19 on power_sequence adds READY qualification -> LIGHT_ENABLE_SAFE.
MCU_EN reset remains here; U18 reaches it through current-limiting R42.



POWER SEQUENCE

SUPPLY QUALIFICATION / HARDWARE LIGHTING INHIBIT



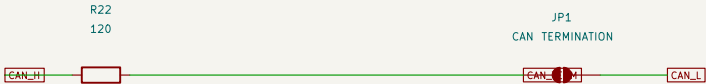
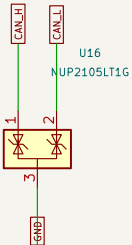
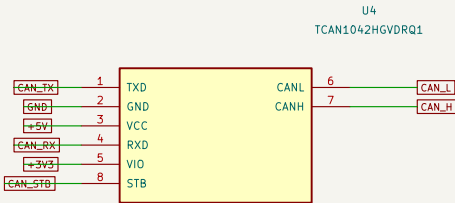
U18 monitors +3V3: 3.07 V nominal falling threshold.
CT intentionally open: 12–28 ms reset release delay.
R42 limits discharge from existing MCU_EN capacitor C1.
MCU_EN pullup R1, capacitor C1 and manual S1 are on controller.

READY also feeds USB R41/R39; independent of ARM.
Prototype timing and 0<V3V3<1.65 V behavior require measurement.
After reset: initialize outputs OFF, verify PCA state, require fresh enable.

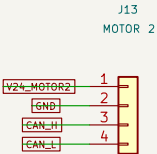
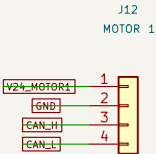
ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /power_sequence/		
File: power_sequence.kicad_sch		
Title: Power Sequence		
Size: A3	Date: 2026–09–06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 3/19

CAN

CAN: 5V transceiver supply / 3.3V logic interface.
Exactly two endpoint terminations: close JP1 only if this controller is an endpoint.

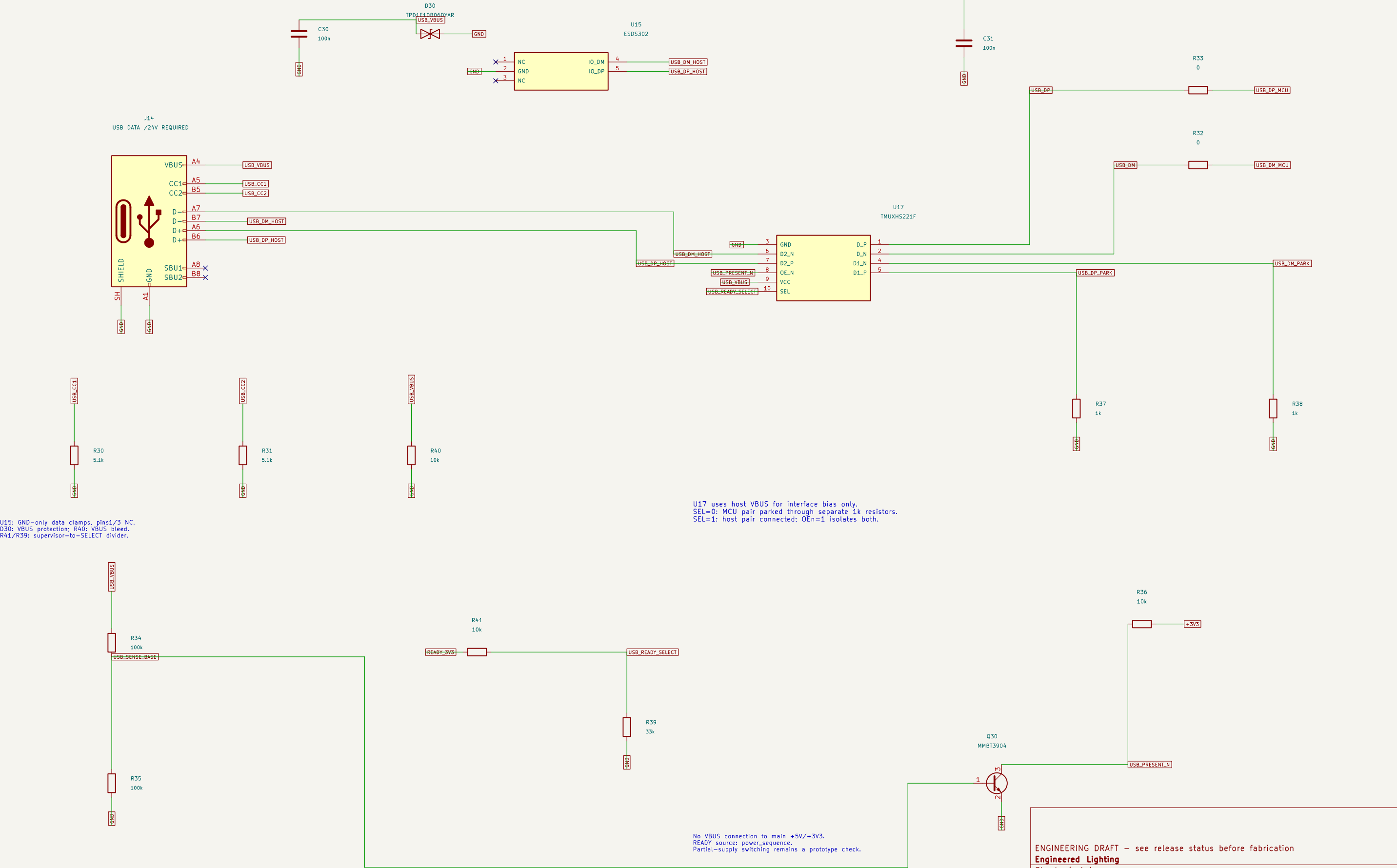


Motor connector: 1 = fused +24V, 2 = GND, 3 = CANH, 4 = CANL.
Unit-specific motor harness polarity must be verified before connection.



USB

USB DATA / MAIN BOARD REQUIRES EXTERNAL 24 V



U15: GND-only data clamps, pins1/3 NC.
D30: VBUS protection; R40: VBUS bleed.
R41/R39: supervisor-to-SELECT divider.

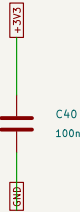
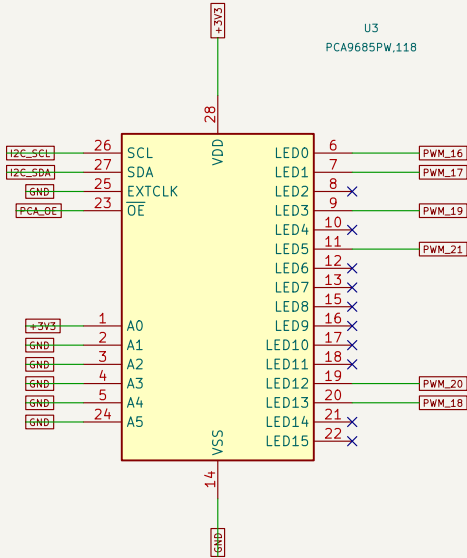
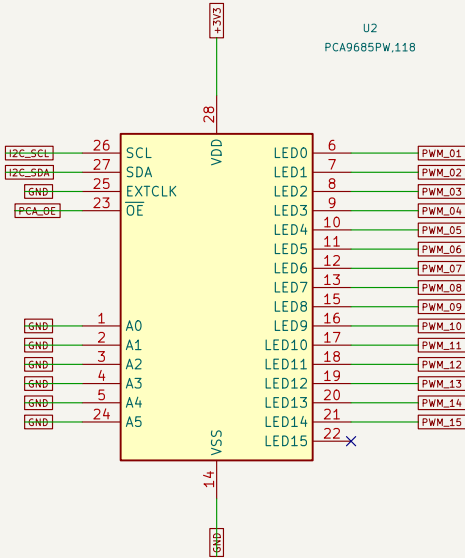
U17 uses host VBUS for interface bias only.
SEL=0: MCU pair parked through separate 1k resistors.
SEL=1: host pair connected; OE=1 isolates both.

No VBUS connection to main +5V/+3V3.
READY source: power_sequence.
Partial-supply switching remains a prototype check.

PWM

U2 address 0x40 / zones 1–5

U3 address 0x41 / zones 6–7



24 outputs total: 21 here + 3 direct spotlight PWM.
OUTDRV=0, OUTNE=00; INVRT/duty encoding must be verified with outputs inhibited.

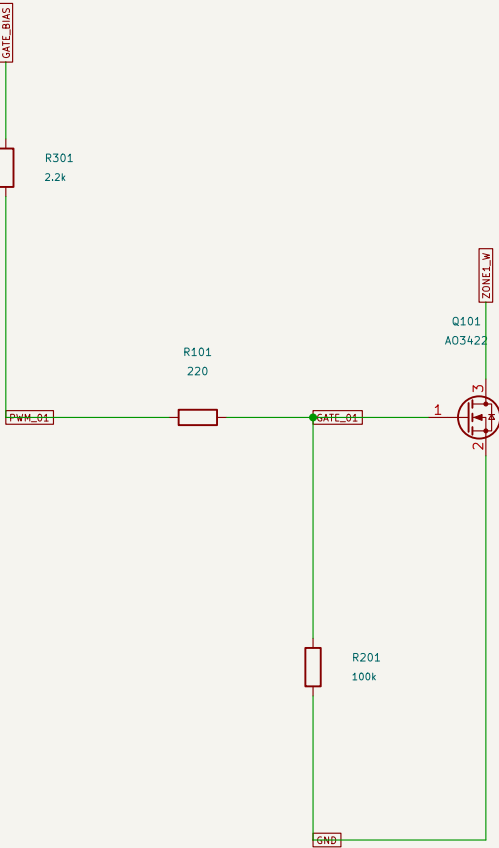
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Engineered Lighting
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Title: Pwm		
Size: A3	Date: 2026–09–06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 6/19

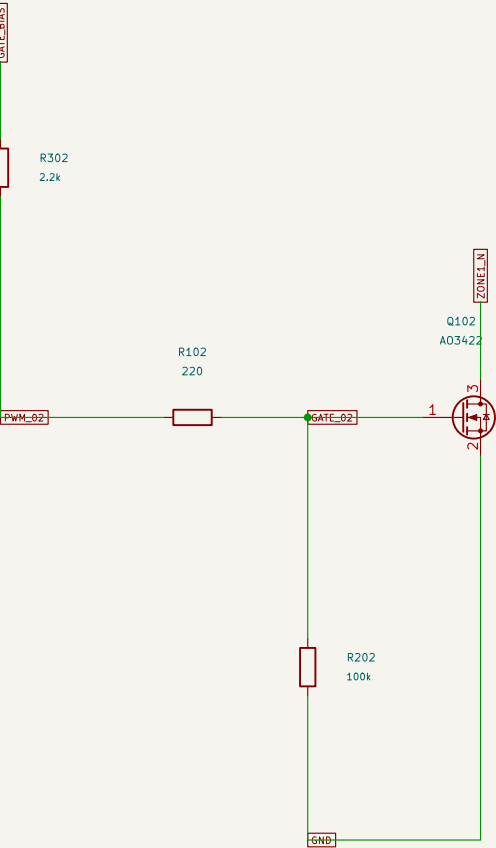
ZONE 1

24 V common positive; three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

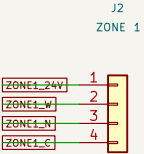
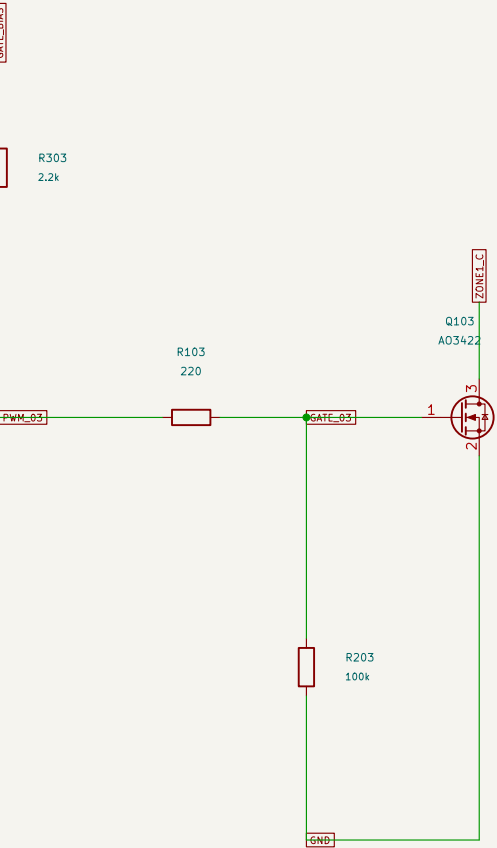
WARM / channel 1



NEUTRAL / channel 2



COOL / channel 3



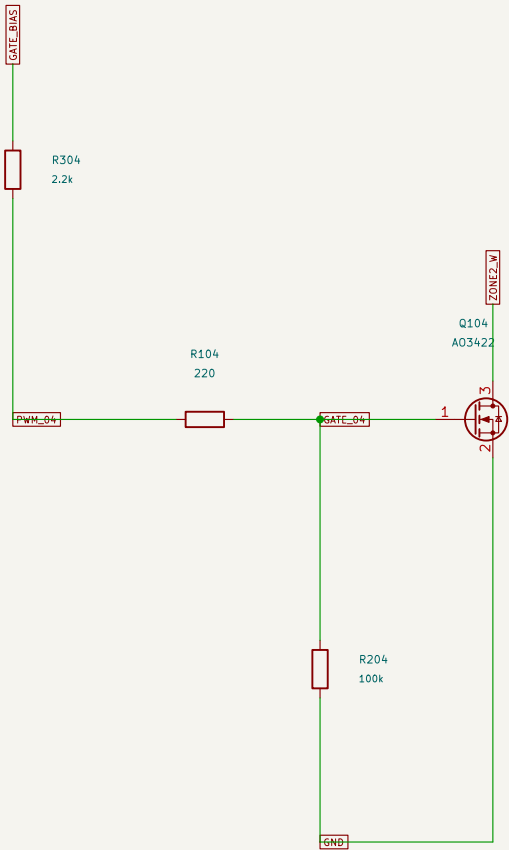
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_1/ File: zone_1.kicad_sch		
Title: Zone 1		
Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 7/19

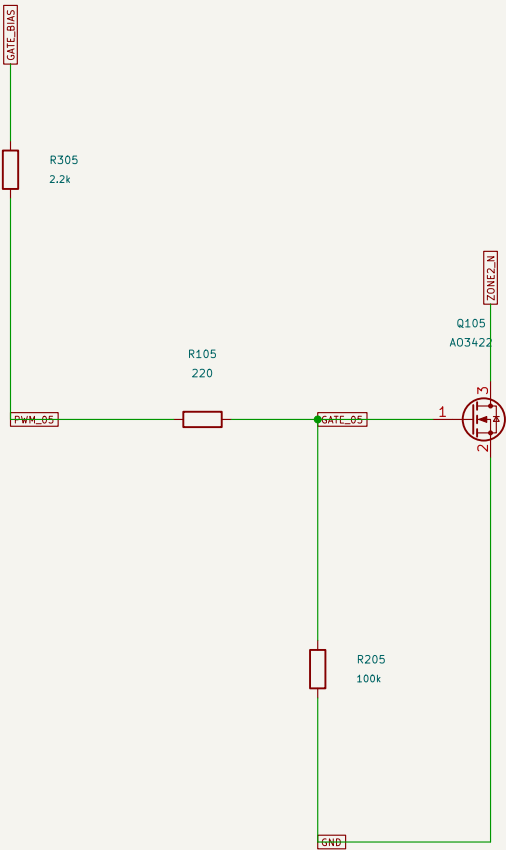
ZONE 2

24 V common positive; three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

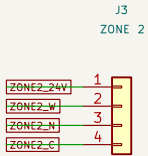
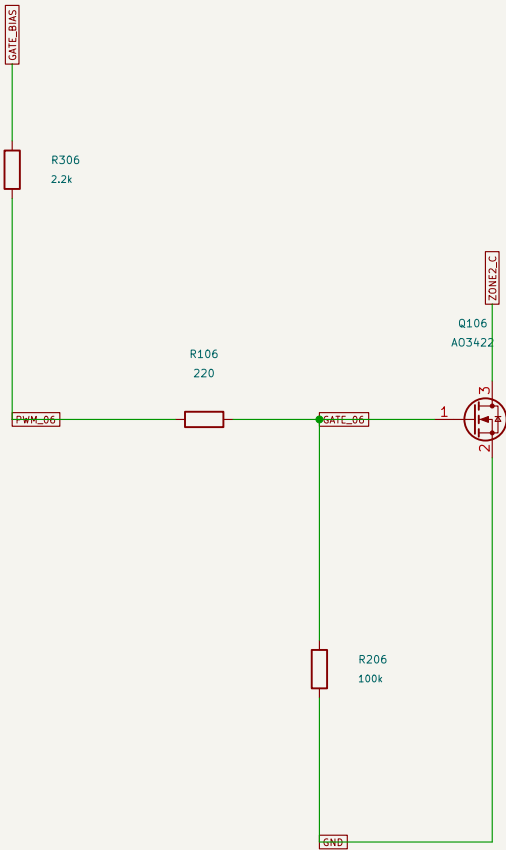
WARM / channel 4



NEUTRAL / channel 5



COOL / channel 6



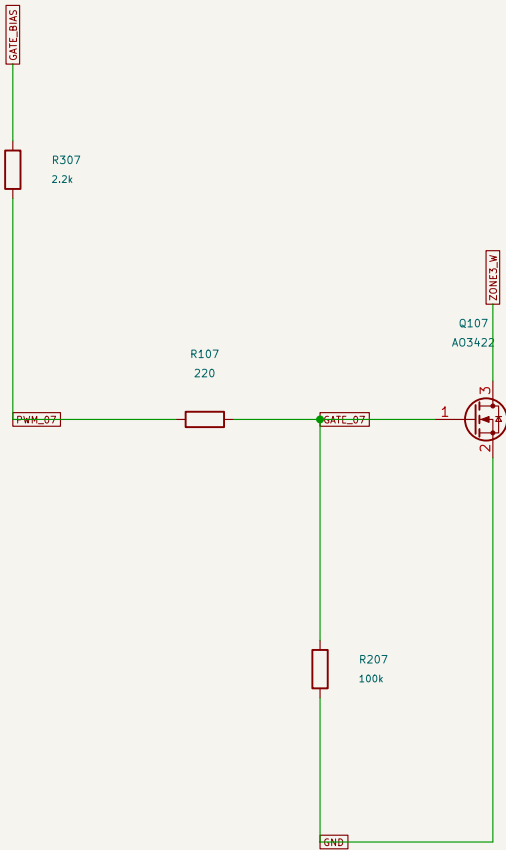
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_2/ File: zone_2.kicad_sch		
Title: Zone 2		
Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 8/19

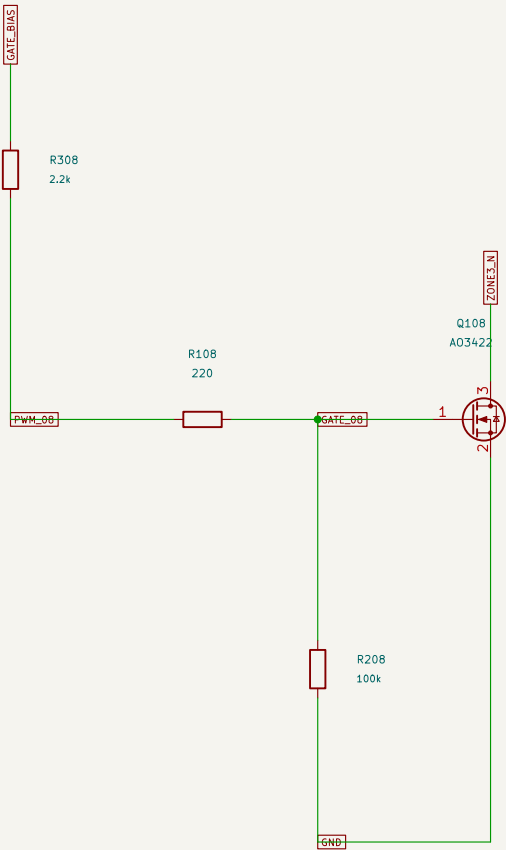
ZONE 3

24 V common positive; three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

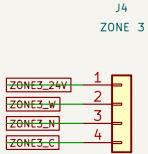
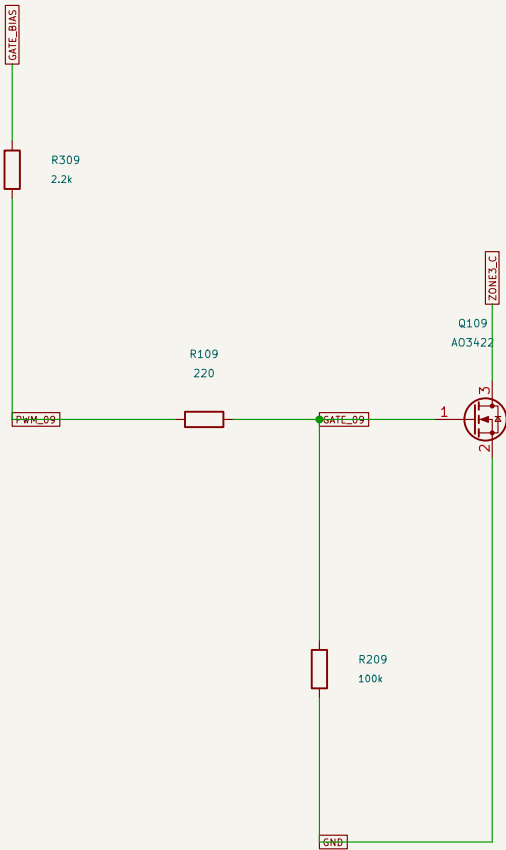
WARM / channel 7



NEUTRAL / channel 8



COOL / channel 9



GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_3/ File: zone_3.kicad_sch		
Title: Zone 3		
Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 9/19

ZONE 5

24 V common positive; three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

WARM / channel 13

NEUTRAL / channel 14

COOL / channel 15

J6 ZONE 5

ZONE5-24V	1
ZONE5-W	2
ZONE5-N	3
ZONE5-C	4

GATE_BIAS

R313 2.2k

PWM_13

R113 220

GATE_13

Q113 AO3422

ZONE5_W

R213 100k

GND

GATE_BIAS

R314 2.2k

PWM_14

R114 220

GATE_14

Q114 AO3422

ZONE5_N

R214 100k

GND

GATE_BIAS

R315 2.2k

PWM_15

R115 220

GATE_15

Q115 AO3422

ZONE5_C

R215 100k

GND

GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication
Engineered Lighting
Sheet: /zone_5/
File: zone_5.kicad_sch
Title: Zone 5

Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3	Id: 11/19	

ZONE 5

24 V common positive: three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

WARM / channel 13

NEUTRAL / channel 14

COOL / channel 15

J6
ZONE 5

1
2
3
4

ZONE5-24V
ZONE5-W
ZONE5-N
ZONE5-C

GATE_BIAS

R313
2.2k

R113
220

PWM-13

GATE-13

R213
100k

GND

Q113
AO3422

ZONE5-W

1
2
3

GATE_BIAS

R314
2.2k

R114
220

PWM-14

GATE-14

R214
100k

GND

Q114
AO3422

ZONE5-N

1
2
3

GATE_BIAS

R315
2.2k

R115
220

PWM-15

GATE-15

R215
100k

GND

Q115
AO3422

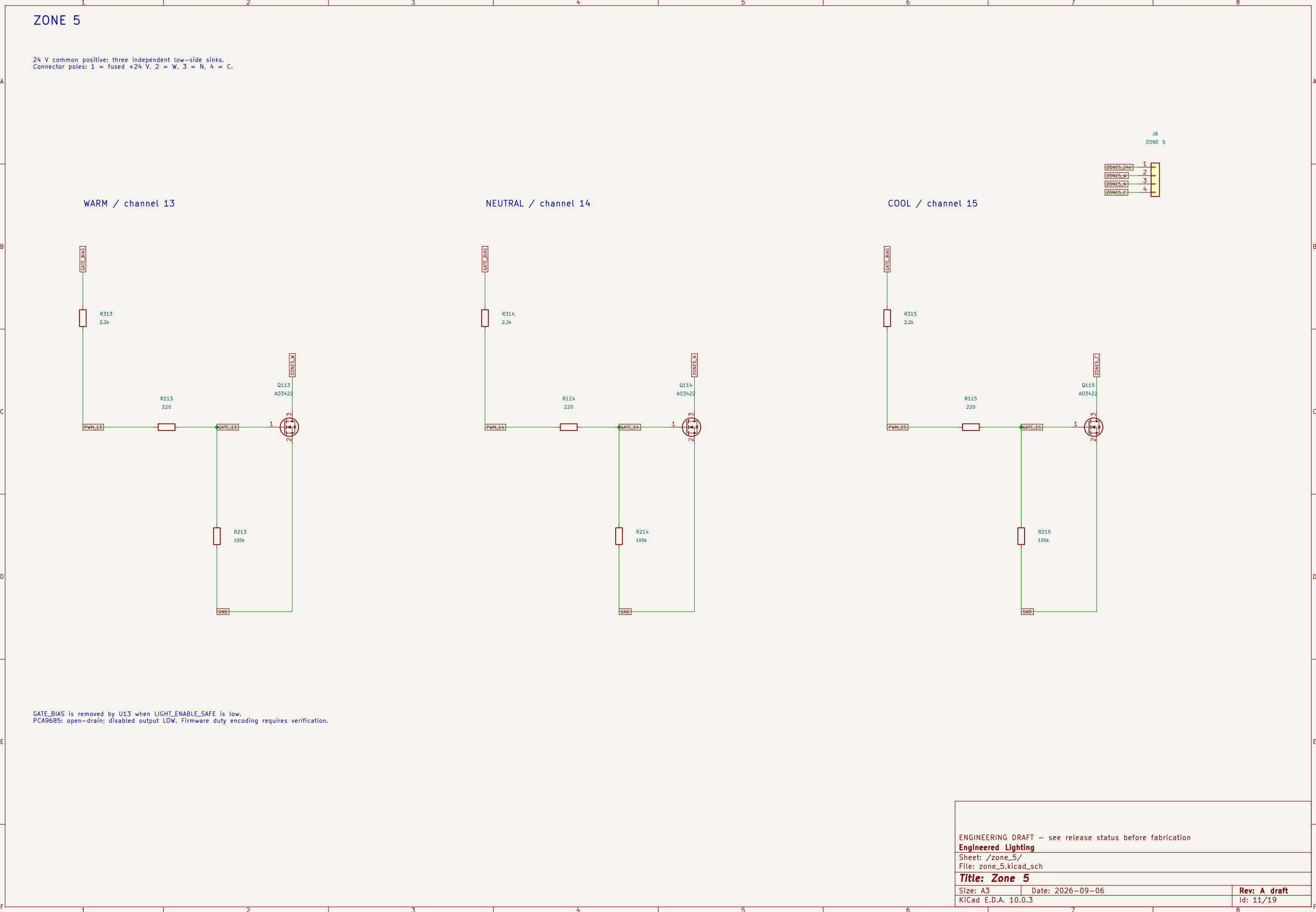
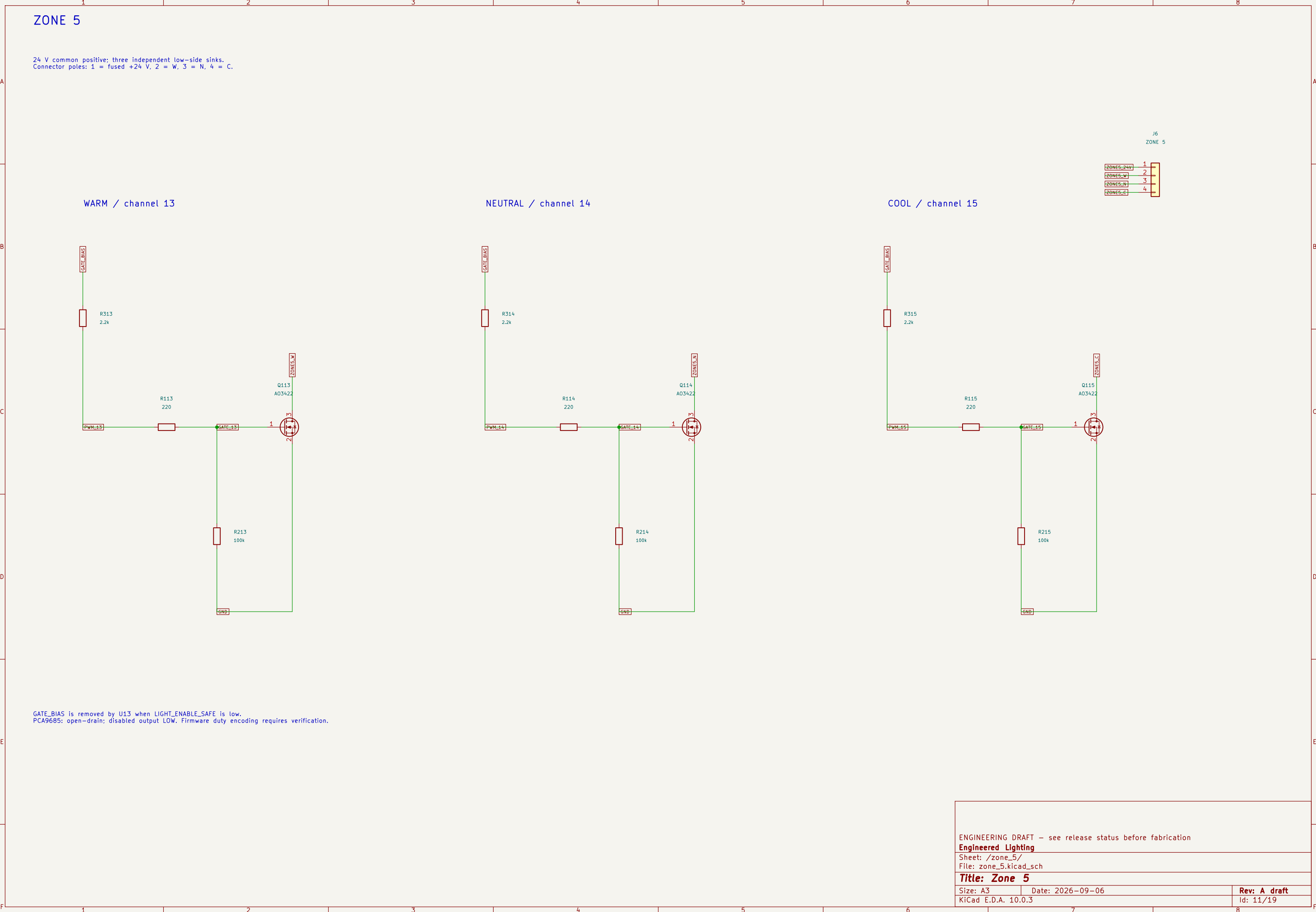
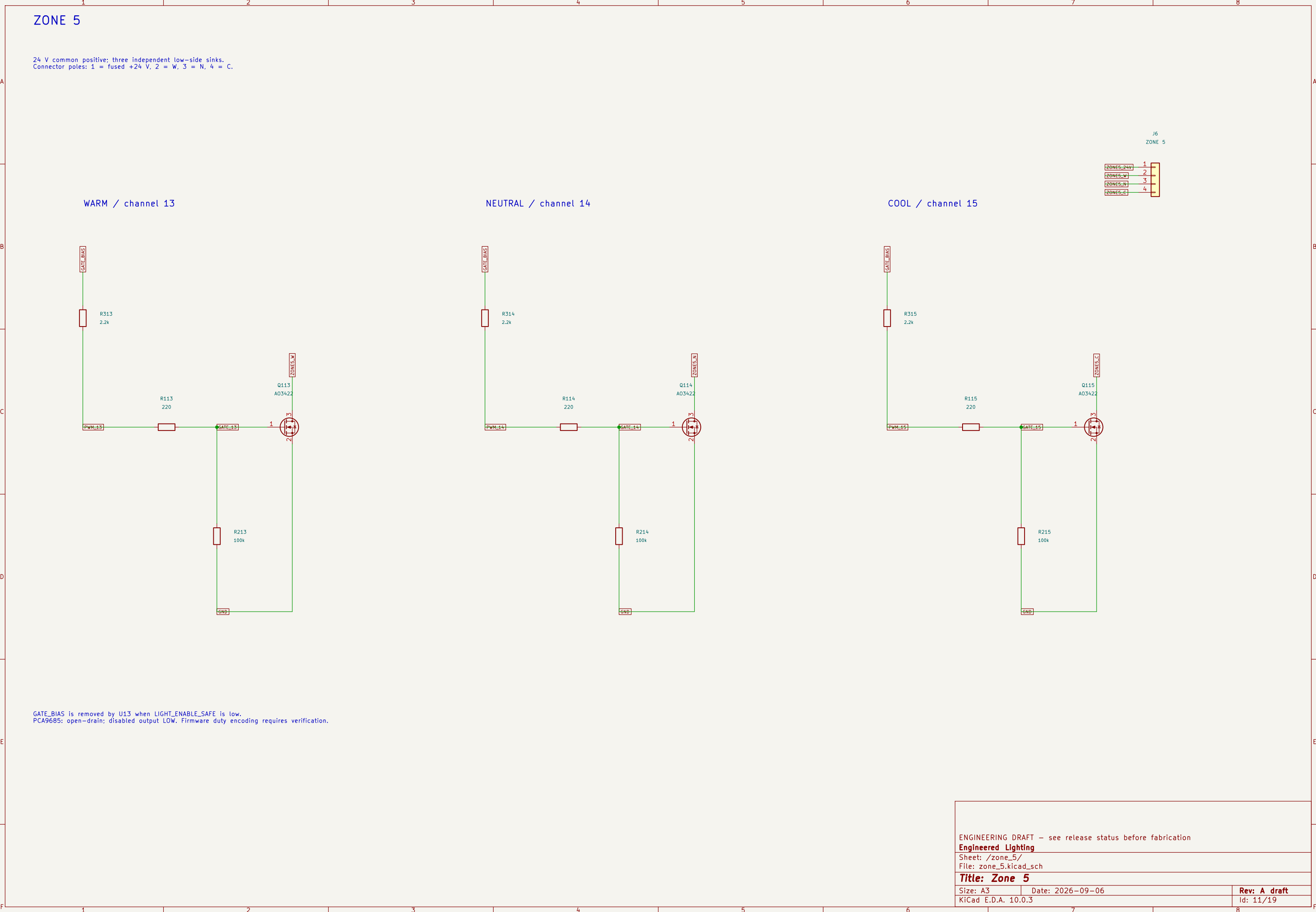
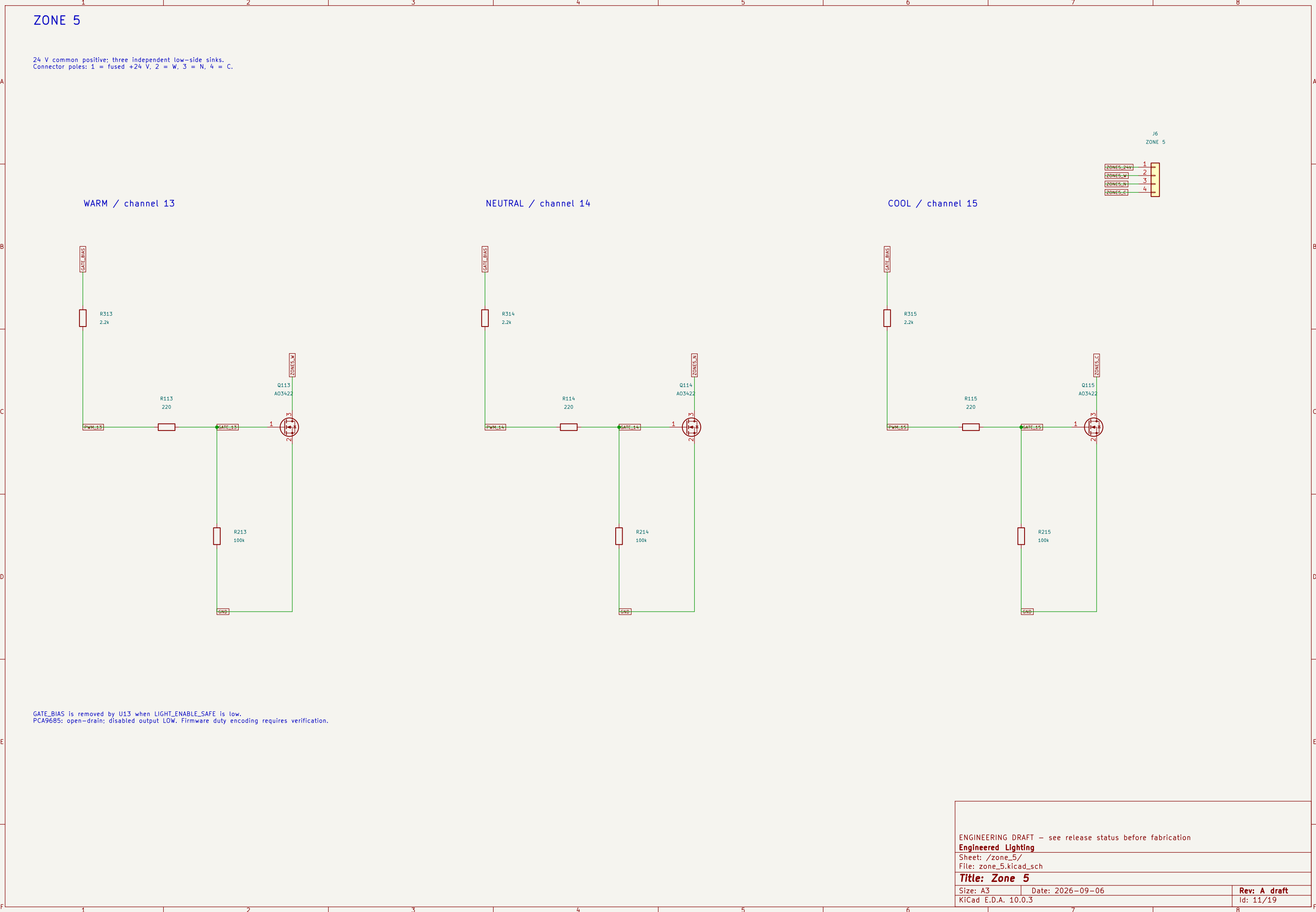
ZONE5-C

1
2
3

GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication
Engineered Lighting
Sheet: /zone_5/
File: zone_5.kicad_sch
Title: Zone 5

Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 11/19



ZONE 5

24 V common positive: three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

WARM / channel 13

NEUTRAL / channel 14

COOL / channel 15

J6
ZONE 5

1
2
3
4

ZONE5-24V
ZONE5-W
ZONE5-N
ZONE5-C

GATE_BIAS

R313
2.2k

R113
220

PWM-13

GATE-13

Q113
AO3422

ZONE5-W

1
2
3

R213
100k

GND

GATE_BIAS

R314
2.2k

R114
220

PWM-14

GATE-14

Q114
AO3422

ZONE5-N

1
2
3

R214
100k

GND

GATE_BIAS

R315
2.2k

R115
220

PWM-15

GATE-15

Q115
AO3422

ZONE5-C

1
2
3

R215
100k

GND

GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication
Engineered Lighting
Sheet: /zone_5/
File: zone_5.kicad_sch
Title: Zone 5

Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 11/19

1 2 3 4 5 6 7 8

ZONE 6

24 V common positive: three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

WARM / channel 16

NEUTRAL / channel 17

COOL / channel 18

J7
ZONE 6

1
2
3
4

ZONE6-24V
ZONE6-W
ZONE6-N
ZONE6-C

GATE_BIAS

R316
2.2k

R116
220

PWM-16

GATE-16

Q116
AO3422

ZONE6-W

1
2
3

R216
100k

GND

GATE_BIAS

R317
2.2k

R117
220

PWM-17

GATE-17

Q117
AO3422

ZONE6-N

1
2
3

R217
100k

GND

GATE_BIAS

R318
2.2k

R118
220

PWM-18

GATE-18

Q118
AO3422

ZONE6-C

1
2
3

R218
100k

GND

GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication
Engineered Lighting
Sheet: /zone_6/
File: zone_6.kicad_sch
Title: Zone 6
Size: A3 Date: 2026-09-06 Rev: A draft
KiCad E.D.A. 10.0.3 Id: 12/19

1 2 3 4 5 6 7 8

A

B

C

D

E

F

ZONE 6

24 V common positive: three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

WARM / channel 16

NEUTRAL / channel 17

COOL / channel 18

J7
ZONE 6

1
2
3
4

ZONE6-24V
ZONE6-W
ZONE6-N
ZONE6-C

GATE_BIAS

R316
2.2k

PWM_16

R116
220

GATE_16

Q116
AO3422

ZONE6-W

R216
100k

GND

GATE_BIAS

R317
2.2k

PWM_17

R117
220

GATE_17

Q117
AO3422

ZONE6-N

R217
100k

GND

GATE_BIAS

R318
2.2k

PWM_18

R118
220

GATE_18

Q118
AO3422

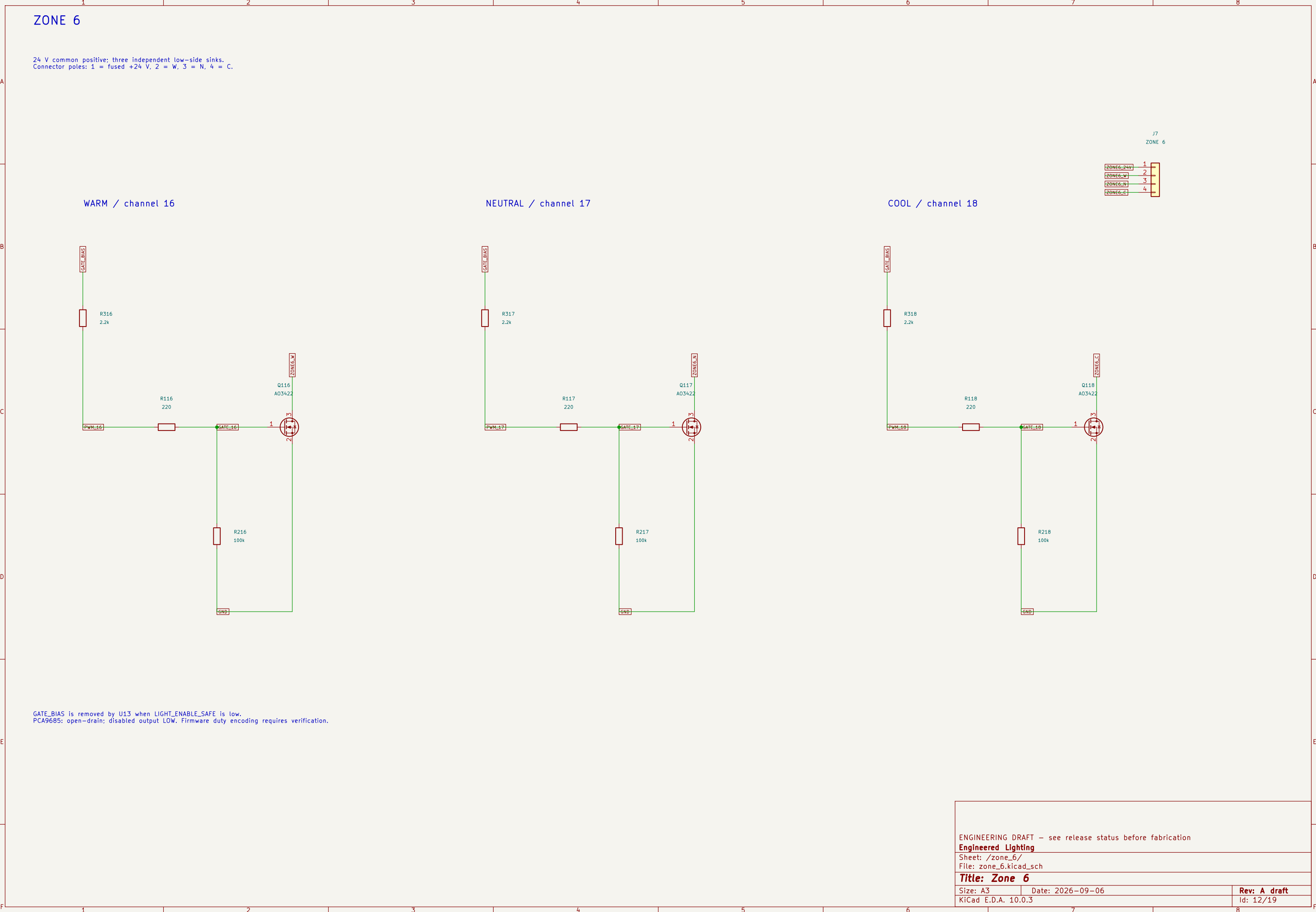
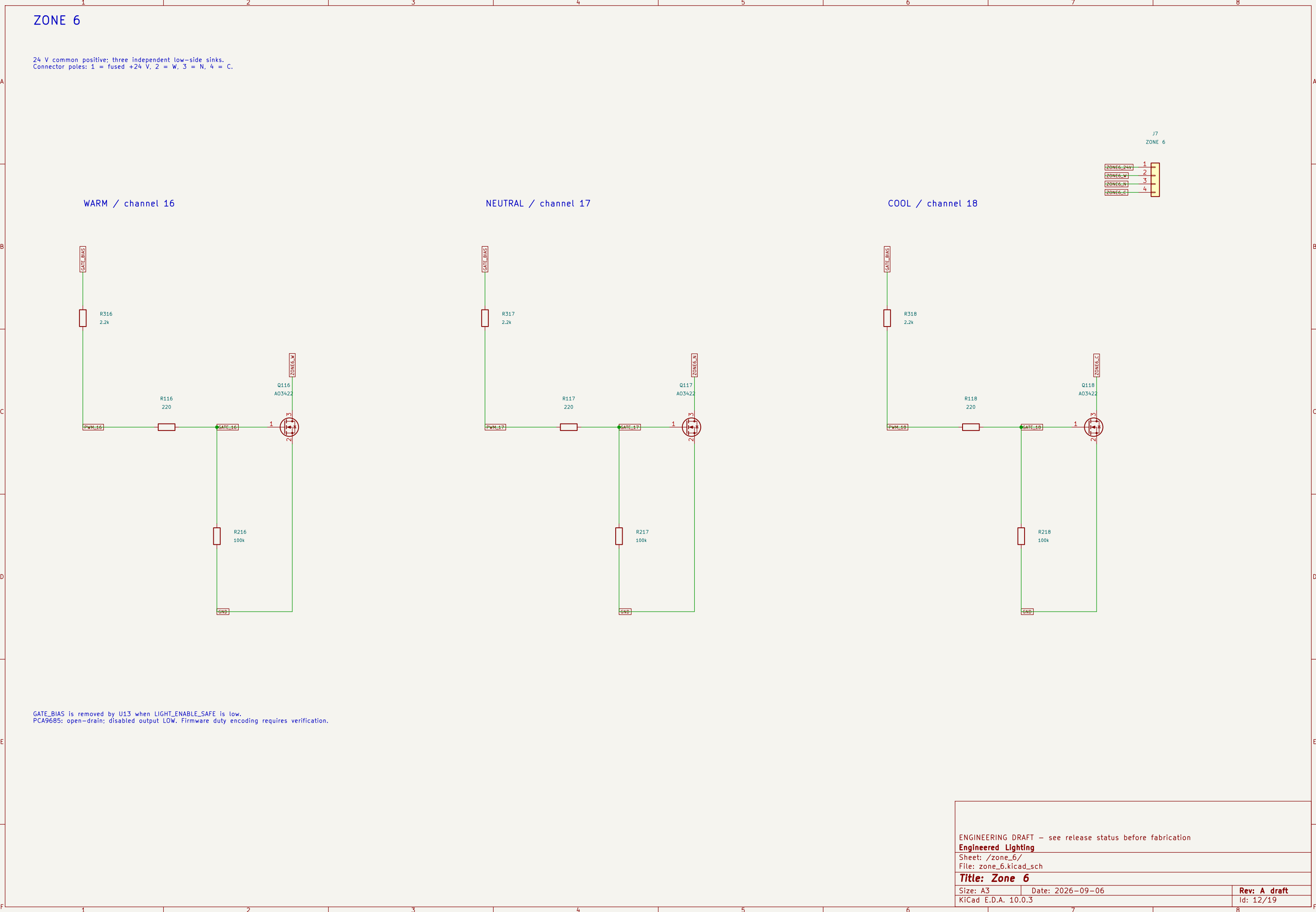
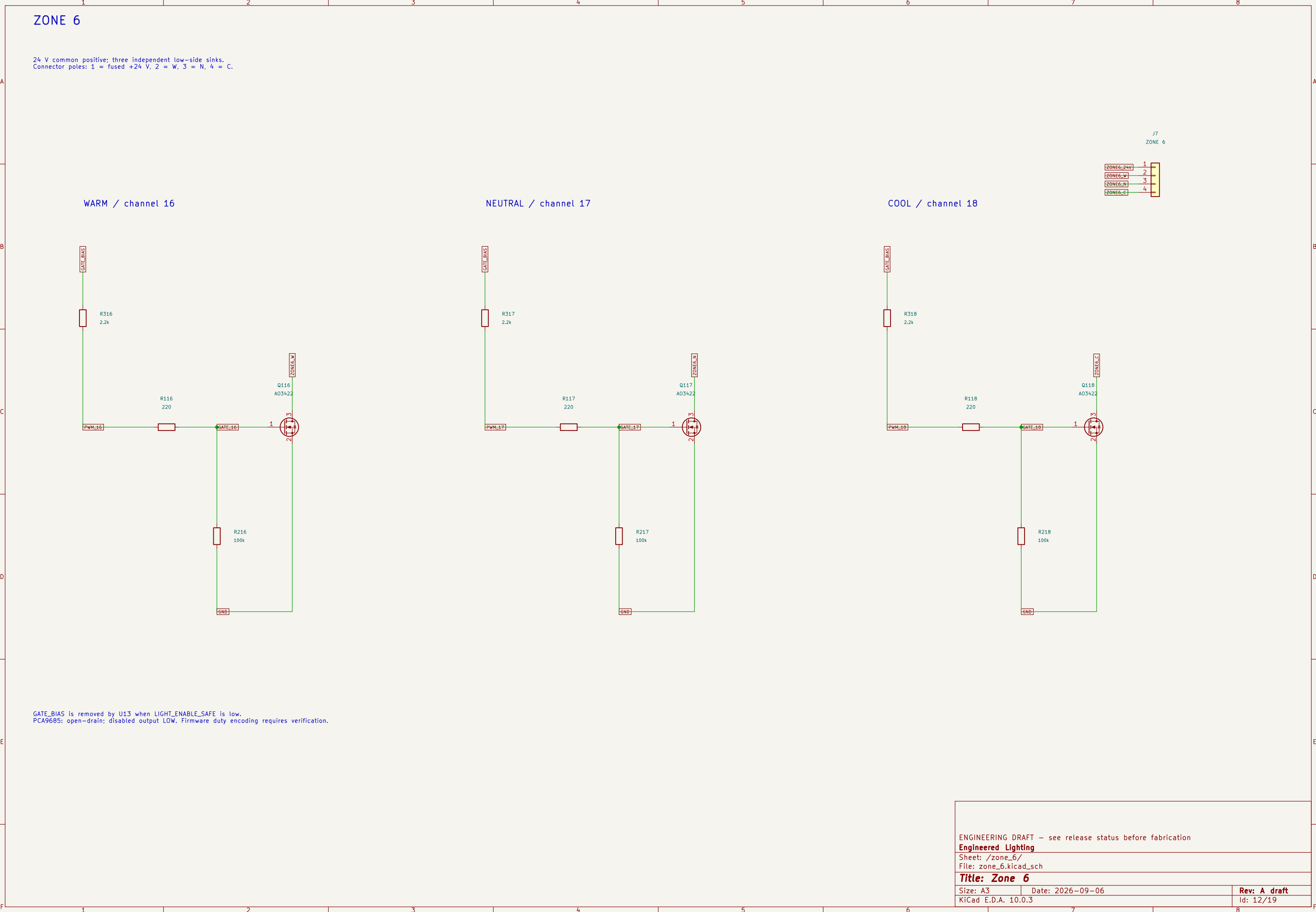
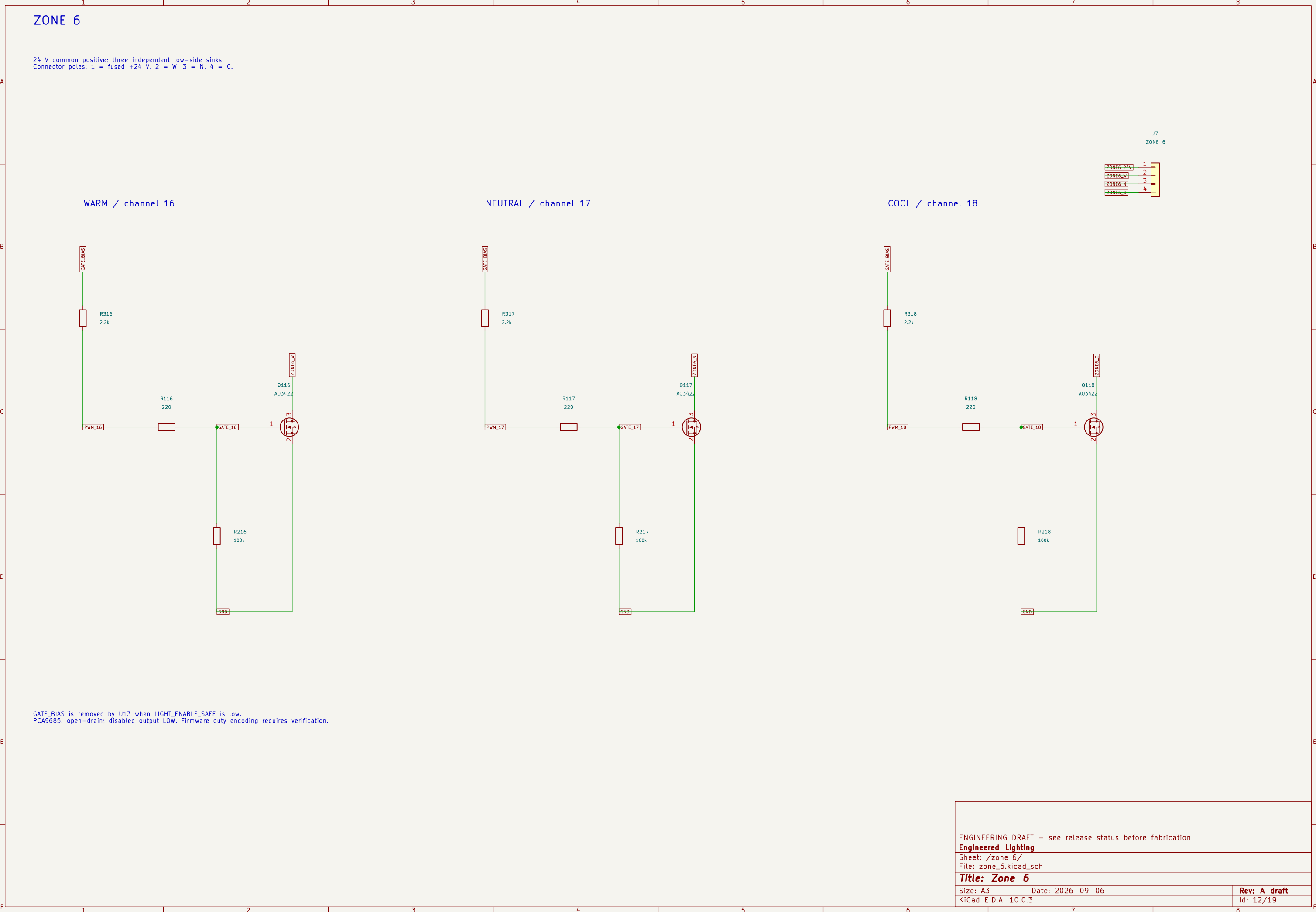
ZONE6-C

R218
100k

GND

GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication
Engineered Lighting
Sheet: /zone_6/
File: zone_6.kicad_sch
Title: Zone 6
Size: A3 Date: 2026-09-06 Rev: A draft
KiCad E.D.A. 10.0.3 Id: 12/19



1 2 3 4 5 6 7 8

A

B

C

D

E

F

ZONE 6

24 V common positive: three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

WARM / channel 16

NEUTRAL / channel 17

COOL / channel 18

J7
ZONE 6

1
2
3
4

ZONE6-24V
ZONE6-W
ZONE6-N
ZONE6-C

GATE_BIAS

R316
2.2k

R116
220

PWM_16

GATE_16

Q116
AO3422

ZONE6-W

R216
100k

GND

GATE_BIAS

R317
2.2k

R117
220

PWM_17

GATE_17

Q117
AO3422

ZONE6-N

R217
100k

GND

GATE_BIAS

R318
2.2k

R118
220

PWM_18

GATE_18

Q118
AO3422

ZONE6-C

R218
100k

GND

GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication
Engineered Lighting
Sheet: /zone_6/
File: zone_6.kicad_sch
Title: Zone 6

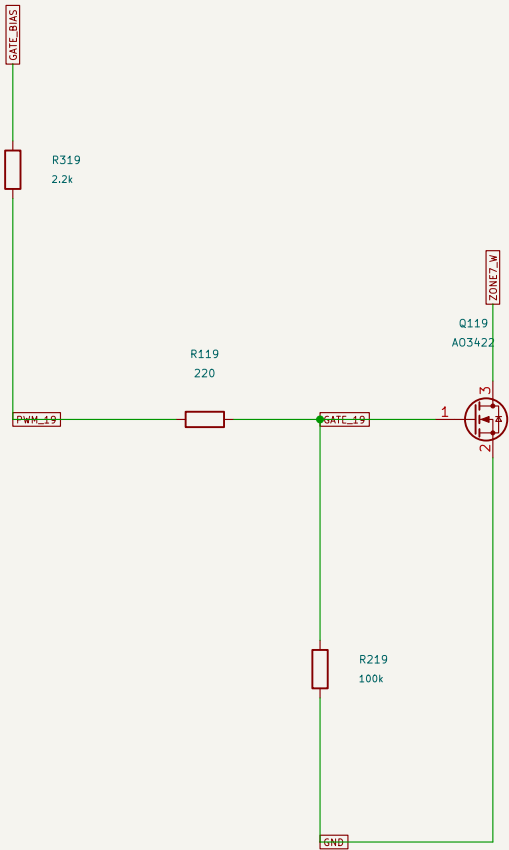
Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 12/19

1 2 3 4 5 6 7 8

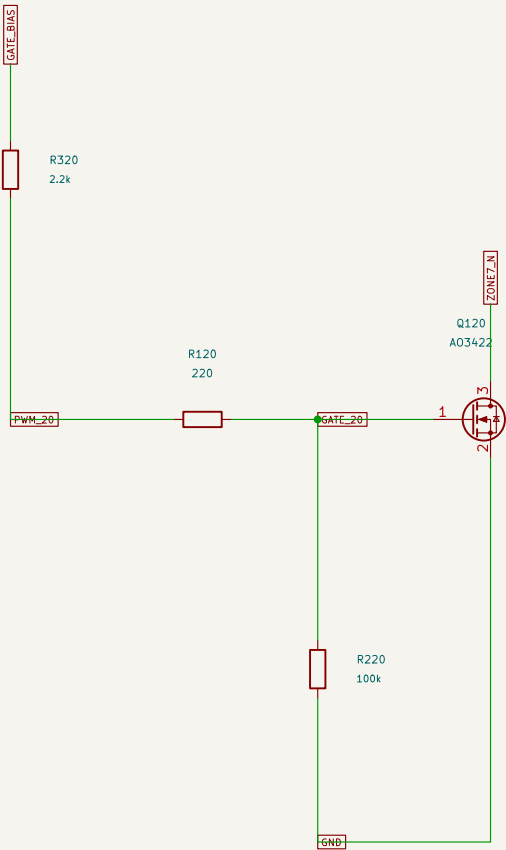
ZONE 7

24 V common positive; three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

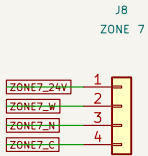
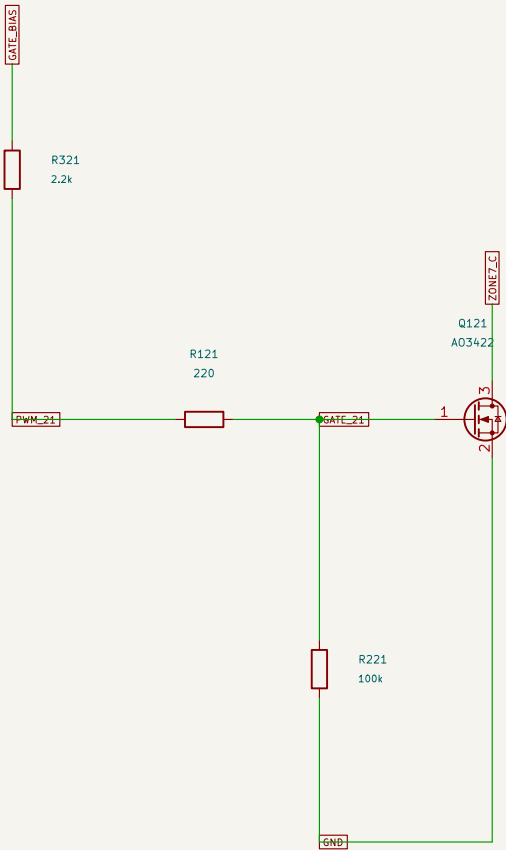
WARM / channel 19



NEUTRAL / channel 20

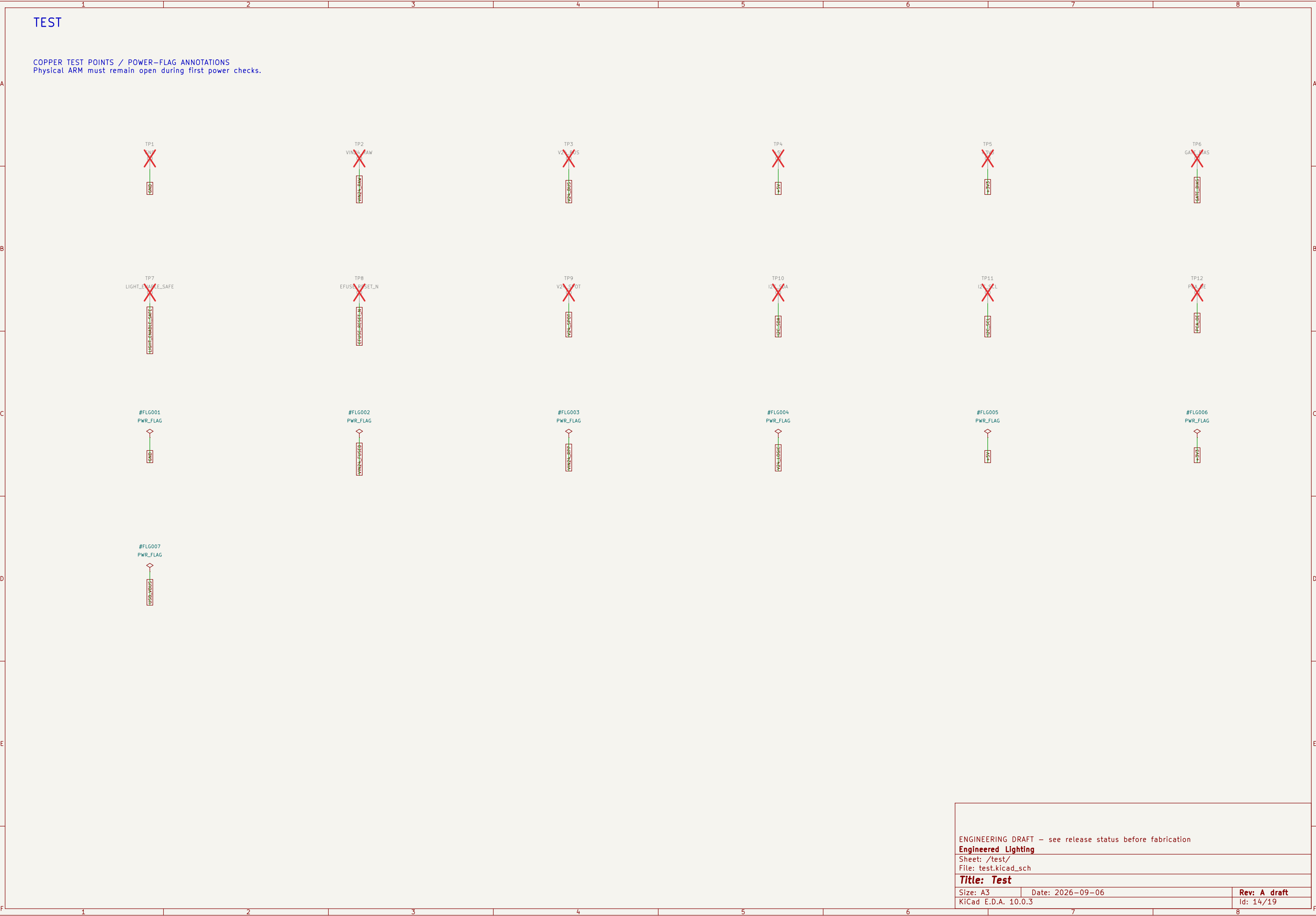


COOL / channel 21



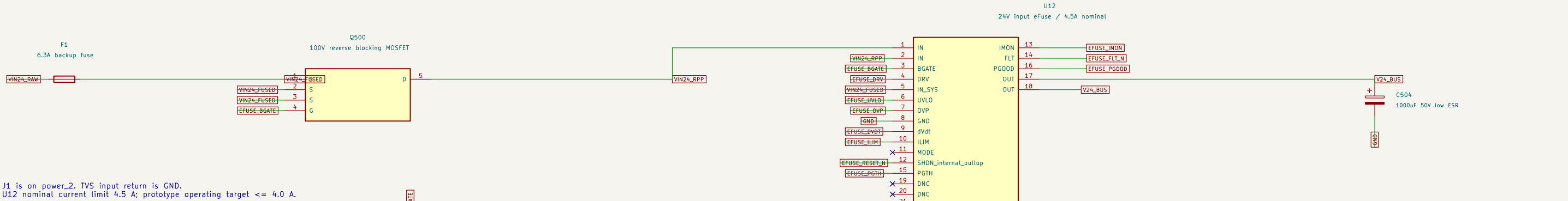
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_7/ File: zone_7.kicad_sch		
Title: Zone 7		
Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 13/19



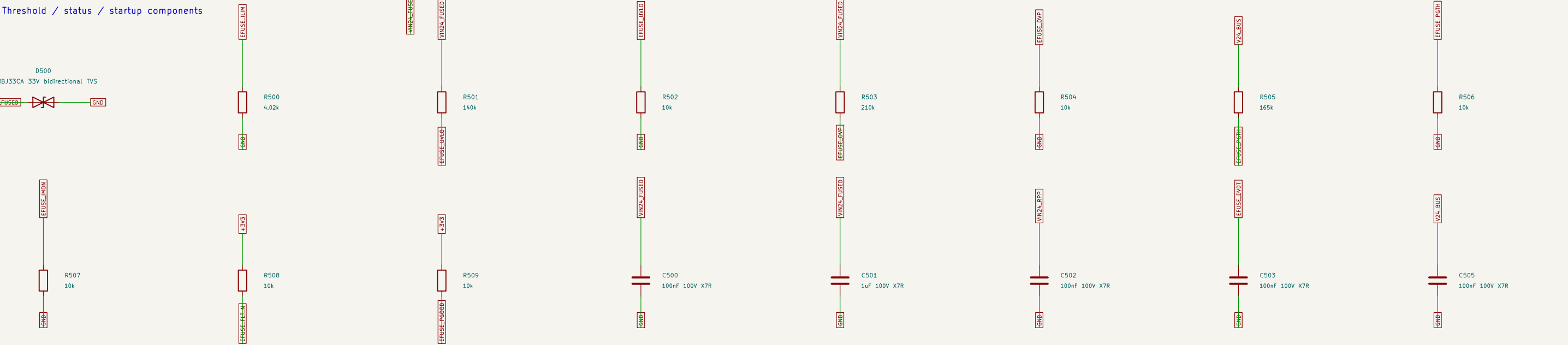
POWER 1

INPUT / REVERSE PROTECTION / MAIN eFUSE



J1 is on power_2. TVS input return is GND.
U12 nominal current limit 4.5 A; prototype operating target <= 4.0 A.

Threshold / status / startup components



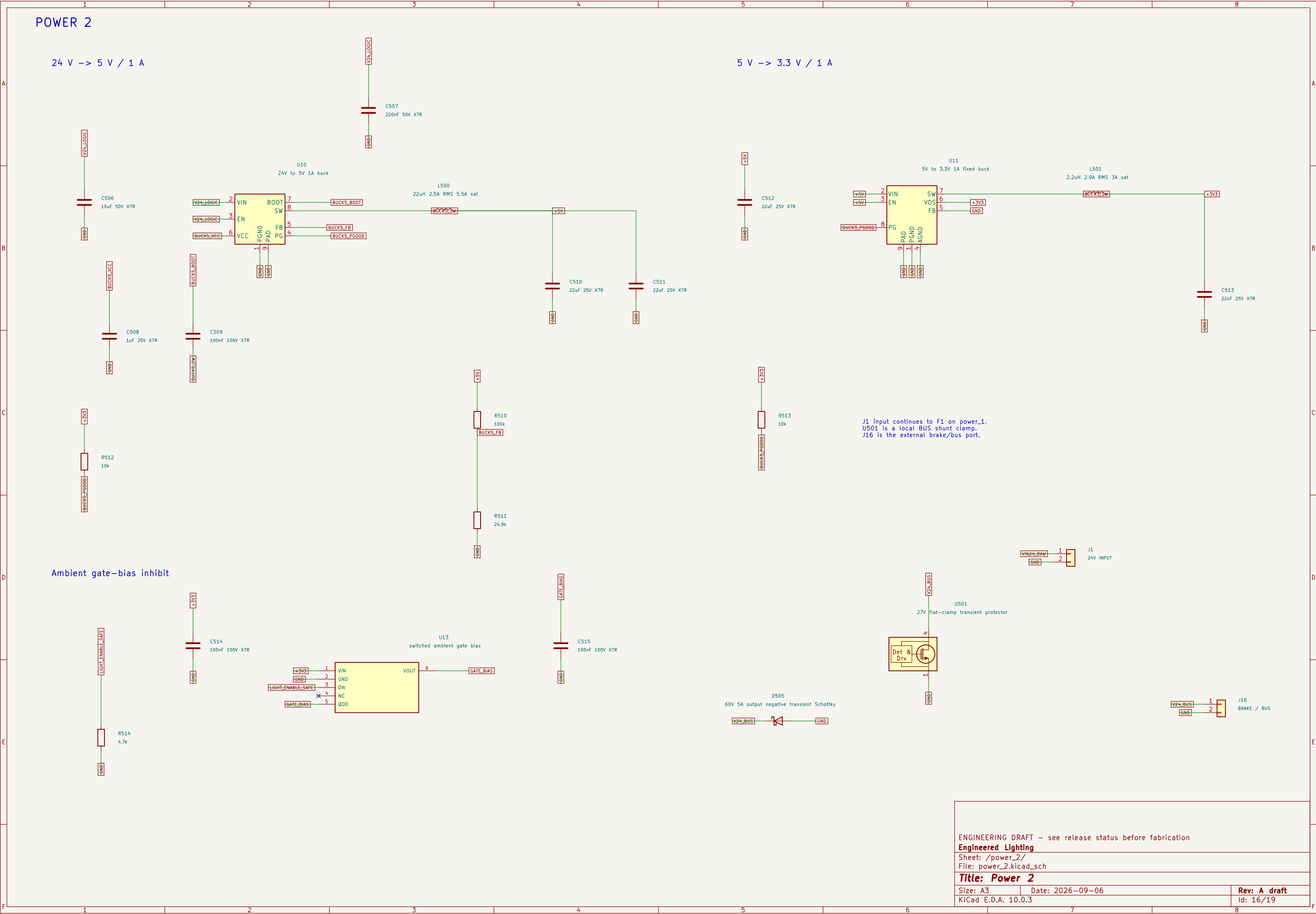
FUSED DISTRIBUTION: motors / ambient zones / logic



POWER 2

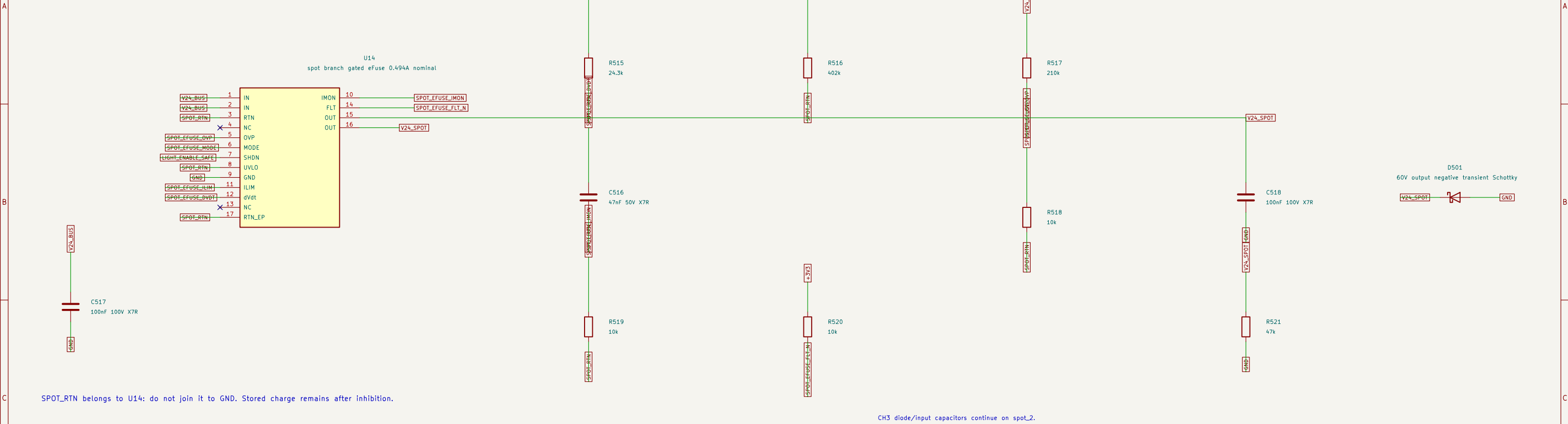
24 V -> 5 V / 1 A

5 V -> 3.3 V / 1 A



SPOT 1

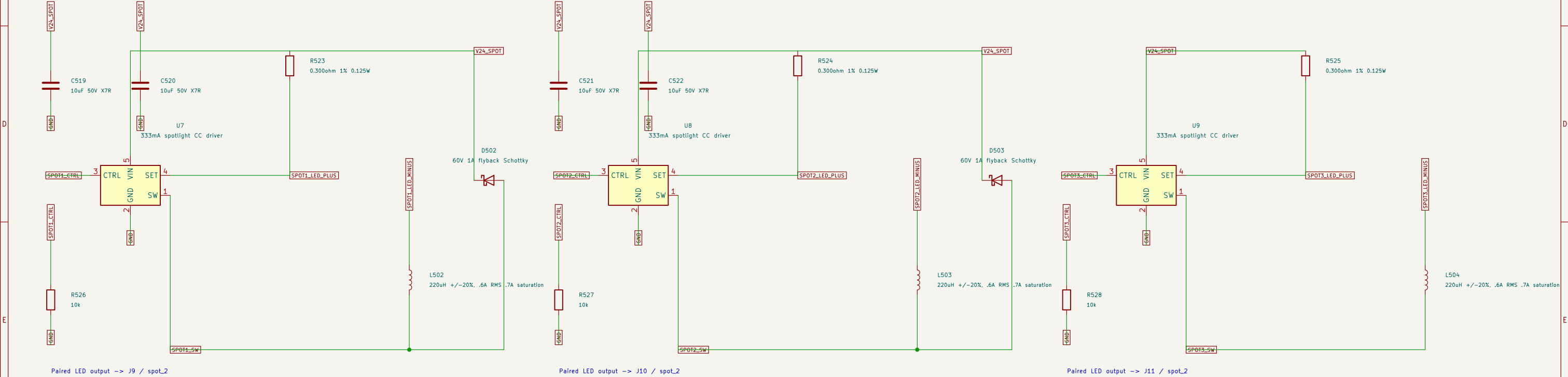
INDEPENDENT SPOT SUPPLY INHIBIT / 0.494 A NOMINAL



SPOT 1 / 333 mA nominal

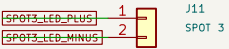
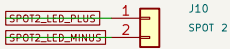
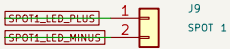
SPOT 2 / 333 mA nominal

SPOT 3 / 333 mA nominal

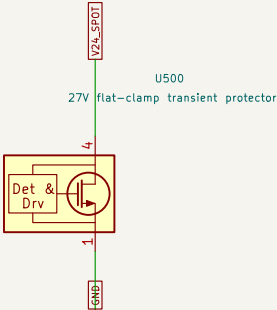
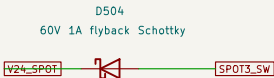
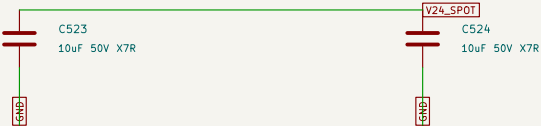


SPOT 2

PAIRED OUTPUTS: pin 1 = LED+, pin 2 = LED-. Neither return is GND.



CH3 current loop continuation from U9 on spot_1



U500 local spotlight-rail shunt clamp.
Pulse limits do not establish continuous braking.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /spot_2/ File: spot_2.kicad_sch		
Title: Spot 2		
Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3	Id: 18/19	

MONITOR EXPANSION

U20
TMP112AIDRLR

6 SDA
1 SCL
4 ADD0
2 GND
5 V+
3 ALERT

H2C-SDA
H2C-SCL
GND

3V3

C46
100n

J18
GPIO EXPANSION 3V3

1 3V3
2 GND
3 GPIO49
4 GPIO20
5 GPIO21
6 GPIO22
7 GPIO23

TP13
GPIO4 STRAP

TP14
GPIO5 STRAP

TP15
GPIO15 STRAP

H1
M3 / 3.2mm NPTH

H2
M3 / 3.2mm NPTH

H3
M3 / 3.2mm NPTH

H4
M3 / 3.2mm NPTH

ENGINEERING DRAFT – see release status before fabrication
Engineered Lighting
 Sheet: /monitor_expansion/
 File: monitor_expansion.kicad_sch
Title: Monitor Expansion
 Size: A3 Date: 2026-09-06 Rev: A draft
 KiCad E.D.A. 10.0.3 Id: 19/19

Engineered Lighting

Title: Monitor Expansion

Title: Monitor Expansion

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Id: 19/19

Id: 19/19